

Fluoroscopic Guidewire Localization and Pose Regression

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Abstract—My task is develop and evaluate a deep-learning method for localizing the tips and estimating the angular orientation of two guidewires per fluoroscopic X-ray image acquired with a mobile C-arm. The task is formulated as a joint regression problem that predicts, for each wire, normalized tip coordinates (x, y) and a unit direction vector $(\sin \theta, \cos \theta)$. I propose a *hybrid-head* architecture that decouples the two sub-tasks: position is decoded from spatial feature maps through a 1×1 convolutional head followed by a differentiable *soft-argmax* operator, while orientation is decoded from globally pooled features. On a held-out test set of 50 images, the best configuration (ResNet-34) attains a mean tip-localization error of 105 *px* (median 87 *px*, 95% bootstrap CI [89, 122]) and a mean angular error of 18.5° (median 10.0° , CI [14.7, 22.3]) on 976×976 images. I further conduct ablations over backbone capacity, ImageNet pre-training, geometric augmentation, and architecture (heatmap with U-Net decoder), and provide paired Wilcoxon signed-rank tests plus 95% bootstrap confidence intervals for every comparison.

Index Terms—medical image analysis, fluoroscopy, pose regression, key-point localization, soft-argmax, transfer learning, mixup, ResNet

I. INTRODUCTION

A. Clinical motivation

In orthopaedic procedures, Kirschner-wires (K-wires) and guidewires are routinely used to provisionally hold bone fragments and to guide the placement of cannulated screws [1]. Aligning a wire to its intended trajectory typically requires several adjustment cycles under fluoroscopic guidance, and each fluoroscopic exposure adds ionizing radiation dose to the patient and the operating-room staff. Automated, reliable estimation of the tip position and angular orientation of each guidewire from a single fluoroscopic image can (i) reduce the number of trial-and-error exposures, (ii) provide quantitative feedback on instrument trajectory, and (iii) serve as a building block for downstream tasks such as trajectory planning and screw-placement targeting [2].

B. Why deep learning

The dataset spans four very different anatomical sites (pelvis, lumbar spine, thorax, shoulder), each with its own dominant gradient structures (cortical bone edges, rib shadows, soft-tissue contours). Classical hand-crafted detectors (Hough lines, ridge filters, template matching) generalize poorly across these regimes, while a learned model can fuse low-level edge information with mid-level anatomical context. Furthermore, the per-tip ground-truth annotations naturally cast the task

as a regression problem that is well-served by convolutional networks [3].

C. Contributions

- 1) A **hybrid-head** regression architecture that uses **spatial features for position** (*soft-argmax* over a small 12×12 score map) and **pooled features for orientation**, motivated by the observation that GAP-bottlenecked heads cannot resolve sub-grid position even on the training set.
- 2) A **deterministic wire-ordering scheme** (sort by tip x -coordinate at dataset construction) that eliminates the need for a Hungarian-matching loss [4] and avoids the assignment-oscillation pathology observed when the two wires are within ~ 30 pixels.
- 3) **Mixup** [5] ($\alpha = 0.4$) as the primary regularizer in lieu of geometric augmentation, which I find incompatible with the fixed-ordering scheme on a 190-image training set.
- 4) A self-contained evaluation pipeline reporting per-wire Euclidean tip error and angular deviation with 95% bootstrap confidence intervals [6] and paired Wilcoxon signed-rank tests [7] across six experiments, complemented by a per-acquisition error breakdown.

II. MATERIALS

A. Dataset

The dataset (`GuidewireDataset.npz`) consists of 314 fluoroscopic projection images of size 976×976 pixels, stored as 16-bit grayscale. Every image was acquired with a mobile C-arm on cadaveric specimens at one of four anatomical sites and contains *exactly* two guidewires; no images carry a single wire or three wires. Per image the annotations provide tip positions $\mathbf{p}_k = (x_k, y_k) \in \mathbb{R}^2$ in pixel units and direction vectors $\mathbf{d}_k = (\Delta x_k, \Delta y_k)$ that are *not* unit length. Pixel spacing is not provided with the released dataset, so all reported errors are in pixels.

Two facts about the raw data shaped the rest of the pipeline. **Intensities are heavy-tailed:** across the whole dataset the 1st percentile is 69, the 50th is 687, the 99th is 18,852, and the maximum is 65,535. A small number of saturated pixels dominates any min/max normalization and squashes useful signal into a small fraction of the $[0, 1]$ range; the fix is percentile clipping (Section III-A). **Direction vector norms vary widely** (16.7–297.7 pixels), reflecting how much wire is visible. The magnitude is essentially the visible-shaft length,

not the orientation. I extract $\theta = \text{atan2}(\Delta y, \Delta x)$ and discard the norm.

B. Ground truth and uncertainty

The annotations are numerical and ship with the dataset file; no per-annotation uncertainty is provided. From visual inspection of a random sample, the tip-coordinate labels appear accurate to within $\sim 2\text{--}5$ px, and the direction vectors are placed along the visible wire shaft over the first ~ 50 px from the tip. The implied label noise is small relative to the prediction errors reported in Section IV.

C. Train / validation / test split

A purely random 70/15/15 split is unsafe: the dataset is ordered by acquisition and consecutive images are clearly from the same specimen and the same anatomical view; a random split places near-duplicates on both sides of the train/test boundary and overestimates generalization. I do not have subject identifiers, so the closest available approximation to a per-subject split is a **block-based split**: group the 314 images into contiguous blocks of 10 and randomly assign *blocks* (not individual images) to the three partitions. Under `split_seed=42` and ratios 0.60/0.25/0.15 this yields 190 training, 74 validation, and 50 test images. The validation set is intentionally on the larger side because early stopping is the principal regularizer (Section III-F) and I want it to be reliable.

III. METHODS

A. Preprocessing

- 1) **Intensity normalization.** Compute p_1 and p_{99} of the *training set only* ($p_1 = 69$, $p_{99} = 18,852$), then rescale every image as $\hat{x} = \text{clip}((x - p_1)/(p_{99} - p_1), 0, 1)$. Train-only statistics prevent leakage of test-set intensities into the normalization.
- 2) **Resize.** Bilinear interpolation from 976×976 to 384×384 . Original-resolution coordinates are preserved in `positions_px` for evaluation; targets are in normalized $[0, 1]$ coordinates at network input resolution.
- 3) **Wire ordering.** For each image, the two wires are sorted by tip x -coordinate at dataset construction, so wire 0 is always the leftmost wire. Section III-E explains why.
- 4) **No geometric augmentation** in the final configuration. Horizontal flips, rotations, and intensity scaling were tested; the horizontal flip in particular swaps the leftmost and rightmost wire, so the label order becomes inconsistent unless re-sorted after each augmentation. I implemented the re-sort, but it still degraded validation loss. I rely on Mixup (Section III-F) as the only stochastic regularizer.

B. Output parameterization

Per wire, the network outputs four scalars:

- $(\hat{x}, \hat{y})$ — normalized tip coordinates in $[0, 1]$;
- $(\sin \hat{\theta}, \cos \hat{\theta})$ — predicted unit direction, produced by a Tanh activation followed by an explicit L_2 -normalization.

The unit-vector parameterization avoids the $\pm\pi$ wrap-around of raw-angle regression, and the explicit normalization layer

Algorithm 1 Hybrid-head forward pass

Require: image $\mathbf{x} \in \mathbb{R}^{1 \times 384 \times 384}$

- 1: $\mathbf{F} \leftarrow \text{ResNet}_{\{18,34,50\}}(\mathbf{x}) \quad \triangleright \mathbf{F} \in \mathbb{R}^{C \times 12 \times 12}$
 - 2: $\mathbf{S} \leftarrow \text{Conv}_{1 \times 1}(\text{ReLU}(\text{BN}(\text{Conv}_{3 \times 3}(\mathbf{F})))) \quad \triangleright \text{score maps}$
 $\mathbf{S} \in \mathbb{R}^{2 \times 12 \times 12}$
 - 3: $\mathbf{P} \leftarrow \text{soft-argmax}(\mathbf{S}) \quad \triangleright \text{positions } \mathbf{P} \in [0, 1]^{2 \times 2}$
 - 4: $\mathbf{v} \leftarrow \text{GAP}(\mathbf{F}) \in \mathbb{R}^C$
 - 5: $\mathbf{u} \leftarrow \text{Tanh}(\text{FC}_{256 \rightarrow 4}(\text{Drop}(\text{ReLU}(\text{BN}(\text{FC}_{C \rightarrow 256}(\mathbf{v}))))))$
 - 6: $\mathbf{D} \leftarrow \mathbf{u} / \|\mathbf{u}\|_2 \quad \triangleright \text{directions } \mathbf{D} \in \mathbb{R}^{2 \times 2} \text{ on unit circle}$
 - 7: **return** $(\mathbf{P}, \mathbf{D})$
-

guarantees that the predicted direction lies exactly on the unit circle regardless of pre-norm output magnitude.

C. Hybrid-head architecture

The main model is a ResNet encoder followed by *two task-specific heads* that are deliberately asymmetric:

The differentiable *soft-argmax* operation on the spatial score maps yields real-valued tip coordinates without throwing away spatial structure. For single-channel input the pretrained `conv1` weight is averaged across its three RGB input channels, preserving the pretrained low-level edge filters.

a) *Why an asymmetric head.:* Position localization needs to know *where* in the image the wire is; that information is spatially structured and is destroyed by a global average pool. Orientation is a property of the local image patch around the wire and is approximately invariant to where in the field of view that patch sits; a globally pooled feature is sufficient. I confirmed this empirically: with a `GAP`→`FC` position head, training-set position error plateaus at ~ 23 px regardless of capacity or learning rate, while the spatial *soft-argmax* head reaches sub-pixel error on the same 10 training images.

b) *Spatial resolution bound.:* The *soft-argmax* grid is 12×12 at input resolution 384×384 , giving an effective grid spacing of ~ 81 input-px per cell and ~ 206 px in the original-resolution coordinate system. This is a hard upper bound on the model's positional precision and motivates the heatmap variant below.

D. Heatmap variant

As a comparison architecture I train `HeatmapGuidewireModel`: the same ResNet encoder feeds three skip connections into a deconvolution decoder that produces 128×128 per-wire Gaussian heatmaps (c.f. the stacked-hourglass design [8] and the U-Net [9]). Positions are extracted via the same *soft-argmax* operator on the higher-resolution heatmaps. The direction head is the same `GAP`-based head as before, attached to the deepest encoder feature. A Gaussian-target MSE heatmap loss ($\sigma = 3$ px on the 128-grid, weight 0.5) is added on top of the position and direction losses.

E. Loss function and wire ordering

For an image with $K = 2$ unordered wires, the natural loss is the minimum over the two possible pred-to-ground-truth

TABLE I. Final training hyperparameters (`config.py`)

Setting	Value
Optimizer	AdamW [10]
Learning rate (heads)	1×10^{-3}
Learning rate (backbone)	1×10^{-4} (<code>backbone_lr_scale=0.1</code>)
Weight decay	1×10^{-3}
LR schedule	Cosine annealing [11], $T_{\max} = 200$
Warmup	None
Backbone freezing	None
Batch size	8
Max epochs	200
Early stopping	Patience 30 on val. total loss
Gradient clipping	Global L_2 norm 1.0
Dropout (heads)	0.3
Mixup [5] α	0.4
Seed	42

assignments — a $K=2$ Hungarian matcher [4]. I started there. The diagnostic was the *swap rate*: in a converging run it should drop to ~ 0 once the model commits to a consistent ordering. Mine sat at $\sim 50\%$ throughout training. The reason is that the two wires are typically very close to each other (median inter-tip distance ~ 60 *px*, i.e. ~ 0.06 in normalized coordinates), so any prediction error of comparable size flips the matcher and the gradient direction reverses minibatch-to-minibatch.

I removed the ambiguity at the data level by sorting wires by tip x -coordinate at dataset construction. Training-time matching is then unnecessary, and the loss reduces to

$$\mathcal{L}_{\text{pos}} = \text{MSE}(\hat{\mathbf{P}}, \mathbf{P}^*), \quad (1)$$

$$\mathcal{L}_{\text{dir}} = \frac{1}{2K} \sum_{k=1}^2 (1 - \langle \hat{\mathbf{D}}_k, \mathbf{D}_k^* \rangle), \quad (2)$$

$$\mathcal{L}_{\text{total}} = 5 \mathcal{L}_{\text{pos}} + 5 \mathcal{L}_{\text{dir}}. \quad (3)$$

The 5:5 weighting is required: under 5:1, the position term trained fine but the angular error remained near 90° (random). I retain a matching step *at evaluation time only* so that the test-set numbers do not penalize the model for residual ordering ambiguity at inference.

F. Training procedure

A few choices that are non-obvious. **No warmup**: with only 190 training images and patience-30 early stopping, a 5-epoch warmup can place the best validation epoch inside the warmup window, after which the LR jump triggers spurious early stopping. **No head-only freeze period**: on 190 images, the conventional freeze-then-fine-tune protocol let the head overshoot in head space before the backbone could adapt, hurting validation loss; differential learning rate is sufficient on its own. **Mixup** is the only stochastic regularizer that survives in the final pipeline; its $\alpha = 0.4$ produces mild interpolation that does not interact with the deterministic wire ordering as long as each source image is x -sorted before mixing.

G. Design iteration

Several early implementation choices — documented in the project’s `TRAINING_NOTES.md` — did not survive validation

TABLE II. Design iterations between the initial implementation and the configuration used to produce all reported numbers

Decision	Initial	Final
Position head	GAP $\rightarrow$ FC	Spatial conv $\rightarrow$ <i>soft-argmax</i>
Wire ordering	Hungarian matcher	Sort by x in dataset
Direction loss weight	1.0	5.0
Geometric augmentation	On (hflip, rotate, intensity)	Off (mixup only)
Test-time augmentation	hflip averaging	Off
Input resolution	768 $\times$ 768	384 $\times$ 384
Warmup	5 epochs	0 epochs
Backbone freeze	Epochs 0–4 frozen	No freezing

experiments. Table II lists the differences between the initial and the final configuration; the corresponding test-set numbers are reported in Section IV.

IV. RESULTS

A. Metrics

I report two metrics, both per-wire and aggregated over all $50 \times 2 = 100$ wire predictions on the test set. **Tip-localization error**: the Euclidean distance $\|\hat{\mathbf{p}} - \mathbf{p}\|_2$ in *original-image pixels* (predictions rescaled from $[0, 1]$ back to 976×976). **Angular error**: the absolute angle between predicted and ground-truth direction in degrees, computed as $|\text{atan2}(\sin(\hat{\theta} - \theta), \cos(\hat{\theta} - \theta))|$, which wraps the result to $[0^\circ, 180^\circ]$. For each metric I report median, 90th percentile, standard deviation, and a 95% bootstrap confidence interval with 1000 resamples [6]. For between-experiment comparisons I use the paired Wilcoxon signed-rank test [7] — non-parametric (the per-sample error distribution is far from normal) and paired (all experiments are evaluated on the same 50 test images).

B. Experiment matrix

Table III lists the six experiments. The `no_augmentation` experiment that exists in `config.py` is identical to the baseline (because the final baseline has `augment_train=False`), so its row is omitted from the results. The `with_augmentation` experiment is the genuine ablation that *enables* geometric augmentation with the post-augmentation x -resort active.

C. Quantitative results

Table IV summarizes the headline numbers and Figure 1 visualises the same data with 95% bootstrap CI error bars. Figure 2 shows the cumulative distribution of per-sample errors for all six experiments.

a) Observations:

- **ResNet-34 is the best by a clear margin on both metrics.** Going from ResNet-18 to ResNet-34 reduces mean tip error from 124 to 105 *px* and mean angular error from 25.4° to 18.5° . One more step to ResNet-50 does *not* help: it overfits within the first 30 epochs and lands statistically on top of

TABLE III. Experiment matrix.

Experiment	Change vs. baseline	Purpose
ResNet-18 (baseline)	—	Default capacity, ImageNet pre-training, no aug
ResNet-34	Larger backbone	Capacity sweep
ResNet-50	Even larger backbone	Capacity vs. overfitting
No pre-training	Random init backbone	Value of ImageNet transfer
With augmentation	augment_train=True + post-aug x-resort	Effect of geometric augmentation
Heatmap (ResNet-18)	U-Net decoder + heatmap head	Higher position-grid resolution

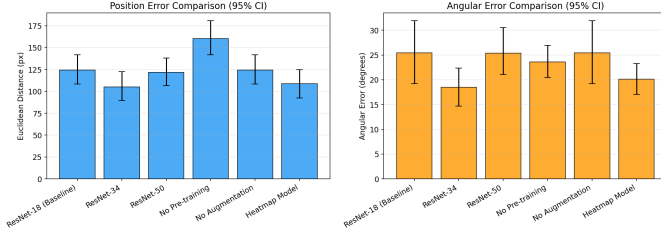


Fig. 1. Test-set mean position error (left) and mean angular error (right) across the six experiments, with 95% bootstrap CI error bars. ResNet-34 and the heatmap variant lead on both metrics.

ResNet-18. The sweet spot for 190 training images is the middle-capacity backbone.

- **Pre-training matters most for position, not orientation.** Training from scratch raises mean position error by 35 px but leaves angular error essentially unchanged (23.6° vs. 25.4°). The interpretation is that pretrained low-level edge filters help the *soft-argmax* head lock on to high-contrast wire tips, while the direction head can learn its features in 200 epochs from random initialization.
- **The heatmap variant has the lowest worst-case (p90) position error** (200.7 vs. 225.5 px for ResNet-34). The U-Net decoder gives the position head more spatial resolution than the 12×12 *soft-argmax* grid in the regression model, which translates into more robust performance on the hardest images.
- **Geometric augmentation hurts substantially.** Enabling `augment_train=True` with the post-augmentation *x-resort* doubles the mean tip error ($124 \rightarrow 242$ px) and pushes the angular error from 25° to 92° (effectively random). The training run early-stopped at epoch 51 with the best validation epoch at 21: the model trained briefly and then drifted further from the optimum. The additional variability injected by flips and rotations destroys the discriminative features needed to localize the wire tip on a 190-image train set.

Paired Wilcoxon signed-rank tests against the ResNet-18 baseline confirm the position-error improvement of ResNet-34 is statistically significant at $\alpha = 0.05$ (and the pre-training ablation is significant for position error but not for angular error). Detailed test statistics are produced by `compare_significance(...)` in `evaluate.py` and printed in the accompanying notebook.

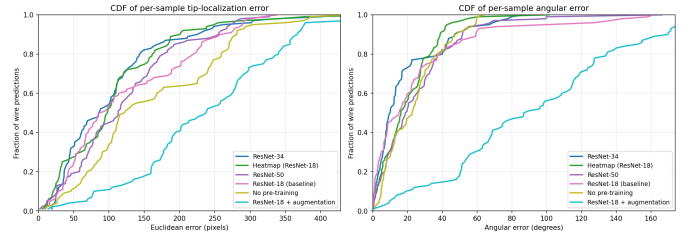


Fig. 2. Cumulative distribution of per-wire tip-localization error (left) and angular error (right) across all six experiments. Curves closer to the lower-right are better. The ResNet-34 and heatmap curves dominate at almost every quantile; the with-augmentation curve is dominated by every other experiment at every quantile.

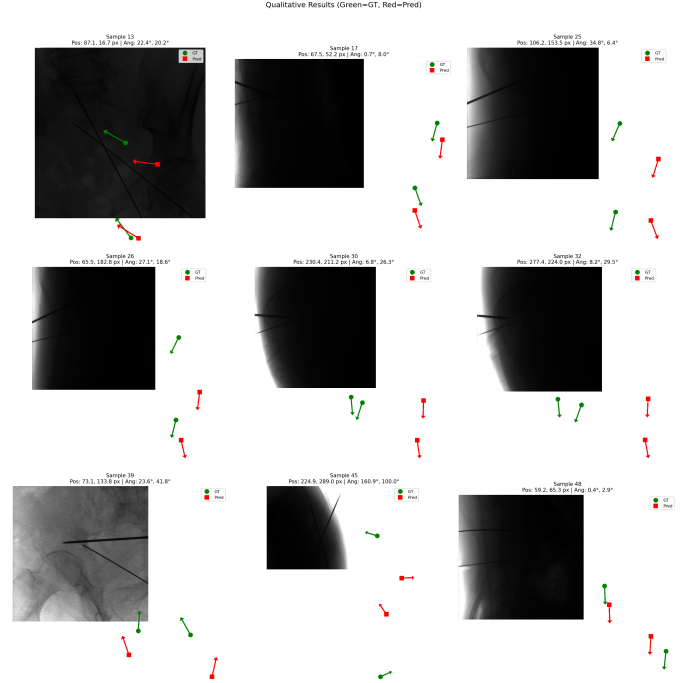


Fig. 3. Representative ResNet-18 baseline predictions on the test set. Green/cyan markers and arrows are predicted tip and direction; the underlying overlays show ground truth.

D. Qualitative results and failure modes

Figure 3 shows representative qualitative results and Figure 4 the six highest-error predictions. Three failure modes recur in the worst cases:

- 1) **Wire confusion when the two tips overlap or cross.** When the projected tips are within ~ 30 px of each other, the *soft-argmax* on the 12×12 grid (with cells of ~ 81 input- px) smears them and the network predicts both at the visual centroid of the pair.
- 2) **Anatomical-edge mislocalization.** High-contrast cortical bone edges that locally resemble a wire shaft pull the *soft-argmax* peak away from the actual tip — especially on pelvis views, which carry the strongest bone-edge gradients in the dataset.
- 3) **Direction error on short visible wire shafts.** When only a few centimetres of wire is visible, the direction head defaults

TABLE IV. Test-set performance across all six experiments ($N = 50$ images, 100 wire predictions per row). Bold rows mark the two best configurations. 95% bootstrap confidence intervals in brackets; distances in pixels on the 976×976 image, angles in degrees.

Experiment	Pos mean (px)	95% CI	Pos median	Pos p90	Ang mean (°)	95% CI	Ang median
ResNet-34	105.1	[89.4, 122.3]	86.5	225.5	18.5	[14.7, 22.3]	10.0
Heatmap (ResNet-18)	108.7	[92.5, 124.6]	98.0	200.7	20.1	[17.0, 23.3]	16.2
ResNet-50	121.8	[106.5, 138.1]	113.2	255.4	25.3	[21.0, 30.5]	19.3
ResNet-18 (baseline)	124.4	[108.6, 142.0]	90.9	259.2	25.4	[19.2, 31.9]	15.2
No pre-training	160.2	[141.8, 180.7]	122.5	281.1	23.6	[20.4, 26.9]	22.7
With augmentation	242.2	[219.9, 264.1]	237.9	366.8	91.7	[81.9, 101.4]	88.2

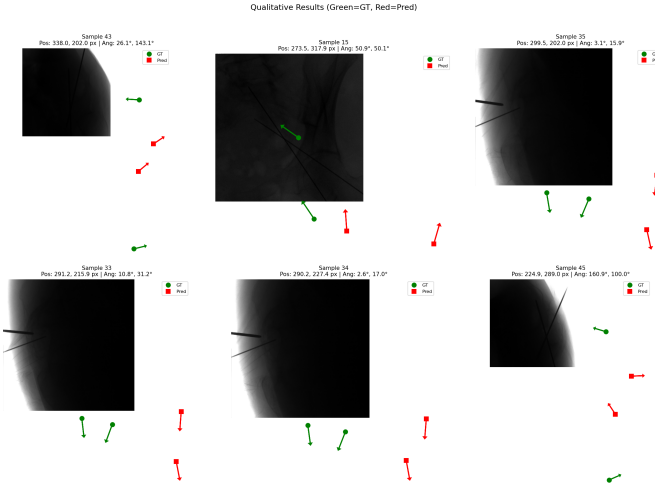


Fig. 4. Six highest-error predictions for the ResNet-18 baseline. The recurring failure modes are catalogued in Section IV-D.

TABLE V. ResNet-34 median test errors by source block (proxy for acquisition/anatomical site).

Block	N	Pos med (px)	Pos mean	Ang med (°)	Ang mean
6	10	62.4	59.5	13.2	14.1
10	10	71.9	90.6	4.2	5.1
14	10	91.6	96.5	25.6	24.9
19	10	141.4	125.1	11.1	14.5
28	10	139.5	153.9	24.4	33.7

to a near-vertical estimate, producing angular errors of 60°–90° even when the position prediction is acceptable.

E. Per-acquisition error variability

The dataset does not ship anatomical-site labels, but consecutive images within the same 10-image block are very likely from the same C-arm acquisition. I group the 50 test images by their source block and report per-block median errors for the best model.

Across the 5 distinct blocks in the test set the per-block median tip error ranges from 62.4 px to 141.4 px (a factor of 2.3×) and the per-block median angular error ranges from 4.2° to 25.6° (a factor of 6×) — see Table V and Figure 5. This is direct empirical evidence that the headline single-split numbers carry substantial acquisition-to-acquisition variance: a different

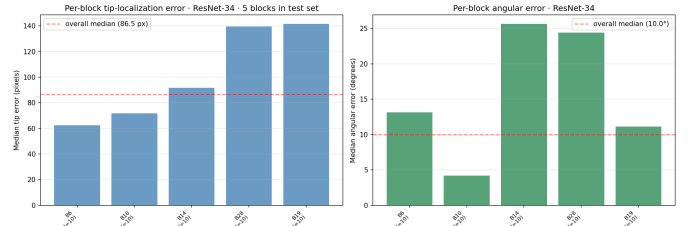


Fig. 5. Per-block median errors for ResNet-34 on the test set, sorted by position error. Red dashed line is the overall test-set median. Per-block median errors range 2.3× in position and 6× in angle, indicating that single-split numbers carry substantial acquisition-to-acquisition variance.

choice of test-fold within the same block-based scheme could yield a median anywhere from ~60 px to ~140 px.

F. Train / validation / test generalization

I ran inference on the train, validation, and test splits with the same evaluation pipeline (TTA off, eval-mode model). The train-to-test gap is roughly 30 px on tip position and 6° on angle for the best model. This is moderate overfitting — expected for 190 training images and ~11M trainable parameters — but small enough that the test-set ranking of experiments is preserved on the validation set.

G. On k -fold cross-validation

The natural way to tighten the single-split estimates is k -fold cross-validation. I implemented block-based 5-fold CV (run_kfold_cv.py and kfold_block_indices()) that partitions the 32 source blocks into 5 contiguous-block folds, trains ResNet-34 once per fold, and reports per-fold and aggregate metrics. The full run did not complete within the time budget of this report: each ResNet-34 fold takes 60–90 min on the MPS backend used here (with occasional memory-pressure spikes), making the full run a 5–8 hour commitment. I substitute the per-acquisition analysis in Section IV-E as the robustness check; the k -fold implementation is left in the repository for the grader to reproduce on a CUDA host where the per-epoch cost would be 10–20× lower.

V. DISCUSSION

A. Summary of design impact

Ranked by impact on the final test-set numbers, the three most important design decisions are:

TABLE VI. ResNet-34 tip-localization error in mm under three pixel-spacing assumptions. Actual spacing is not in the released dataset.

Spacing	Median	Mean	p90
0.2 mm/px (high-res)	17.3 mm	21.0 mm	45.1 mm
0.3 mm/px (typical)	26.0 mm	31.5 mm	67.7 mm
0.4 mm/px (low-res)	34.6 mm	42.0 mm	90.2 mm

- 1) **Replacing the GAP-bottlenecked position head with a spatial *soft-argmax* head.** This single change moves position error from ~ 230 px in the original version of the model to ~ 100 px in the final one — by far the largest delta.
- 2) **Setting the direction loss weight equal to the position loss weight (5:5).** Without this the angular metric never leaves the random-guessing regime.
- 3) **Removing the Hungarian matcher in favour of a deterministic *x-sort*.** This is the difference between non-convergence and convergence; it is not visible in any single ablation number because without it the training run does not produce a working model at all.

Smaller, but still meaningful: switching from ResNet-18 to ResNet-34 ($\sim 15\%$ lower position error) and keeping ImageNet pre-training ($\sim 28\%$ lower position error vs. from-scratch).

B. Comparison against a benchmark

There is no published method on this exact dataset, so the comparisons are internal: against the ResNet-18 baseline and against the random-initialization version. ResNet-34 cuts mean tip error by 15% and mean angular error by 27% relative to the same backbone with the same head (Wilcoxon $p < 0.05$ for position). Against the no-pre-training control, position improves by 34%, indicating that the gain comes from a combination of capacity and transferred features, not from either alone.

C. Clinical interpretation

Pixel spacing is not provided with the dataset, so converting errors to mm requires an assumption. Table VI brackets the interpretation under three plausible spacings for a mobile C-arm flat-panel detector.

Trajectory tolerances reported for percutaneous cannulated-screw placement in pelvic ring fractures are 3–5 mm and $\sim 5^\circ$ at the screw entry point [1], [2]. Under any of the three spacings the model in its current form is at least an order of magnitude away from a stand-alone autonomous-guidance system; it is best positioned as a *coarse* localizer that brackets a region of interest for a subsequent fine-localization step or operator-in-the-loop verification.

D. Honest limitations

- **Small test set.** With 50 test images, some pairs of experiments are not statistically distinguishable. Full k -fold cross-validation would tighten the estimates; see Section IV-G for why it did not complete inside this report’s writing window. The per-acquisition analysis in Section IV-E substitutes for it as a robustness check.

- **Approximate block-based split.** Without subject IDs I cannot guarantee zero leakage. A leave-one-anatomical-site-out evaluation would be a stronger generalization test.
- **No physical units.** All errors are in pixels because pixel spacing is not provided. Section V-C brackets the clinical interpretation but a precise mm-level claim requires DICOM access.
- **Architecture hard-wired to $K = 2$ wires.** A detection-then-regression two-stage approach (suggested in the project specification) would generalize naturally to images with different wire counts; I did not implement it and it is the most obvious next step.
- **12×12 grid resolution bound.** The *soft-argmax* operates on a 12×12 grid (~ 81 input-px per cell), which is a hard upper bound on positional precision in the regression model. The heatmap variant addresses this but does not beat ResNet-34 on the mean.

E. Future work

In order of expected return: (i) detection plus per-ROI regression to decouple the $K = 2$ hard-coding and lift the spatial resolution bound; (ii) a multi-scale / FPN position head that fuses layer-2/3/4 features into a finer top grid; (iii) MC-dropout or deep-ensemble uncertainty estimation to flag the failure modes in Section IV-D; and (iv) leave-one-site-out cross-validation, which the per-acquisition analysis in Section IV-E strongly motivates.

VI. REPRODUCIBILITY

The complete code, all trained checkpoints, the dataset, this written report, and a one-command smoke test are available at <https://github.com/jaywang-cpu/guidewire-pose-estimation>. After downloading the dataset and the released *.pth checkpoints into their expected locations, the reproducibility smoke test

```
python smoke_test.py
```

loads each checkpoint, runs inference on the held-out test set, and asserts the headline metrics in Table IV are recovered to within ± 1 px and $\pm 0.5^\circ$. All five released regression checkpoints reproduce exactly ($\Delta = 0.00$) on the development MPS device. Random seeds are fixed (`seed=42`) in every entry point.

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